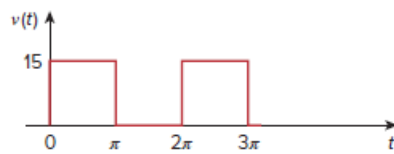


Problem

Find the exponential Fourier series for the waveform in Fig. 17.53.

Figure 17.53:



Step-by-step solution

Step 1 of 3

Consider the following waveform:

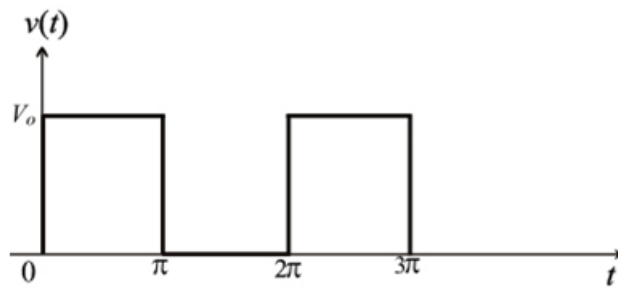


Figure 1

Step 2 of 3

Form the periodic wave form in Figure 1, the time period is $T = 2\pi$.

Calculate the fundamental angular frequency.

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2\pi} \\ &= 1\end{aligned}$$

Write the exponential Fourier series representation for the waveform shown in Figure 1.

$$\begin{aligned}v(t) &= \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \\ &= \sum_{n=-\infty}^{\infty} c_n e^{jn(1)t} \\ &= \sum_{n=-\infty}^{\infty} c_n e^{jnt}\end{aligned}$$

Step 3 of 3

Calculate the exponential Fourier coefficients c_n .

$$\begin{aligned} c_n &= \frac{1}{T} \int_0^T v(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{(2\pi)} \int_0^{2\pi} v(t) e^{-jn(1)t} dt \\ &= \frac{1}{(2\pi)} \left[\left(\int_0^{\pi} (15) e^{-jnt} dt \right) + \left(\int_{\pi}^{2\pi} (0) e^{-jnt} dt \right) \right] \end{aligned}$$

Simplify the equation further.

$$\begin{aligned} c_n &= \frac{15}{2\pi} \frac{1}{-jn} \left(e^{-jnt} \right) \Big|_{t=0}^{\pi} \\ &= \frac{15}{-2jn\pi} (e^{-jn\pi} - 1) \\ &= \frac{j15}{2n\pi} ((\cos n\pi - j \sin n\pi) - 1) \quad (\because \sin n\pi = 0) \\ &= \frac{j15}{2n\pi} (\cos n\pi - 1) \end{aligned}$$

Here

$$\begin{aligned} c_0 &= \frac{1}{T} \int_0^T x(t) dt \\ &= \frac{1}{2\pi} \left\{ \int_0^{\pi} 15 dt + \int_{\pi}^{2\pi} 0 dt \right\} \\ &= \frac{1}{2\pi} \{ 15[t]_0^{\pi} \} \end{aligned}$$

Simplify further.

$$\boxed{c_0 = 7.5}$$

Substitute the expression, $\frac{j15}{2n\pi}(\cos n\pi - 1)$ for c_n in $v(t)$ and 7.5 for c_0 .

$$v(t) = 7.5 + \sum_{n=-\infty}^{\infty} \left[\frac{j15}{2n\pi} (\cos n\pi - 1) \right] e^{jnt}$$

Thus, the exponential Fourier series expression for the waveform $v(t)$ in Figure 1 is

$$\boxed{7.5 + \sum_{n=-\infty}^{\infty} \left[\frac{j15}{2n\pi} (\cos n\pi - 1) \right] e^{jnt}}.$$